

A shellable bipartite graph with an induced six-cycle

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Conjecture 541 asserts an equivalence between shellability of the independence complex of a bipartite graph and chordality. We give a counterexample under both ordinary chordality and chordal bipartiteness.

Construction. Let $v_0, \dots, v_5$ form a six-cycle (indices modulo 6), and attach one new leaf w_i to each v_i . These are all twelve vertices and all twelve edges. Give v_i the color $i \bmod 2$ and w_i the opposite color. This is a proper bipartition. The induced subgraph on the v_i is C_6 , so the graph is neither chordal nor chordal bipartite.

Theorem 1. *The independence complex of this graph is pure and shellable.*

Proof. A maximal independent set contains exactly one of v_i, w_i for every i . It cannot contain both because they are adjacent. If it contained neither, the leaf w_i could be added, contradicting maximality. Thus its facets are exactly

$$F_I = \{v_i : i \in I\} \cup \{w_i : i \notin I\},$$

where I is an independent subset of C_6 . Conversely, each such set is independent and maximal, because the omitted member of each pair is adjacent to the chosen one. Every facet therefore has six vertices. There are $1 + 6 + 9 + 2 = 18$ of them.

Order these facets by increasing $|I|$, breaking ties arbitrarily. Consider an earlier facet F_J and a later facet F_I . They are distinct and $|J| \leq |I|$, so there is an $i \in I \setminus J$. The independent set $I' = I \setminus \{i\}$ has smaller size and therefore its facet occurs earlier. Moreover,

$$F_I \setminus F_{I'} = \{v_i\} \subseteq F_I \setminus F_J.$$

This is the standard facet-difference shelling criterion: for every earlier F_J there is an earlier $F_{I'}$ whose difference from the current facet is one vertex belonging to $F_I \setminus F_J$. See [1], Definition 2.7. Hence the displayed order is a shelling. Since the complex is pure, this argument works for the usual pure definition as well as the general nonpure definition. $\square$

The graph consequently satisfies the bipartite hypothesis and the shellability side of the proposed equivalence, while failing either meaning of chordality relevant to the statement.

Formal verification. Lean uses vertices $0, \dots, 11$, with i representing v_i and $i + 6$ representing w_i . The actual adjacency function is checked to be irreflexive and symmetric. A concrete two-coloring and an injective, induced embedding of C_6 are checked, giving the two failures of chordality from their definitions.

Subsets are encoded as twelve-bit masks. The file proves that every Boolean subset and, classically, every proposition-valued subset has an encoding. It proves that the computable facet criterion (independence and inability to add an outside vertex) is equivalent to genuine inclusion-maximal independence against every subset. All 4096 masks are checked, in 64 blocks, to show that the list of 18 facets is complete and has no repetitions. Every facet is checked to have six vertices. Finally the exact facet-difference criterion used in the proof is verified for every ordered pair of facets. Thus no facet, subset, or required pair is omitted.

The proof uses only `Std` and Lean 4.19.0. Finite checks are kernel-reduced `decide` proofs, with no `native_decide`, unproved mathematical axioms, or admitted steps. The final theorem `conjecture541_false` negates the asserted characterization; the stronger counterexample theorem also excludes chordal bipartiteness.

References

- [1] S. Morey, E. Reyes and R. H. Villarreal, *Cohen–Macaulay, shellable and unmixed clutters with a perfect matching of König type*, J. Pure Appl. Algebra 212 (2008), 1770–1786. doi:10.1016/j.jpaa.2007.11.010.